



Cross-Exchange Limit Order Book Signals for Crypto Prediction

Mengxuan Wei, Molin Yang, and Rongsheng Zhang

Mentor: Hyoeun Lee



Introduction

- i. Motivation
- ii. Research Objective

Motivation



- ❑ Crypto trading is fragmented across exchanges
 - ❑ Four major exchanges: Binance, Bybit, OKX, and Gate.io.
 - ❑ Different order flow
- ❑ Single-exchange views are incomplete
 - ❑ Missing cross-exchange signals (lead-lag effects)
 - ❑ Partial microstructure information (Missing data at some timepoints)
- ❑ Research questions are
 - ❑ Can cross-exchange LOB information improve 5-second return prediction?
 - ❑ Which model performs best under cross-exchange feature integration?
 - ❑ What types of cross-exchange signals are most useful?

Research Objective



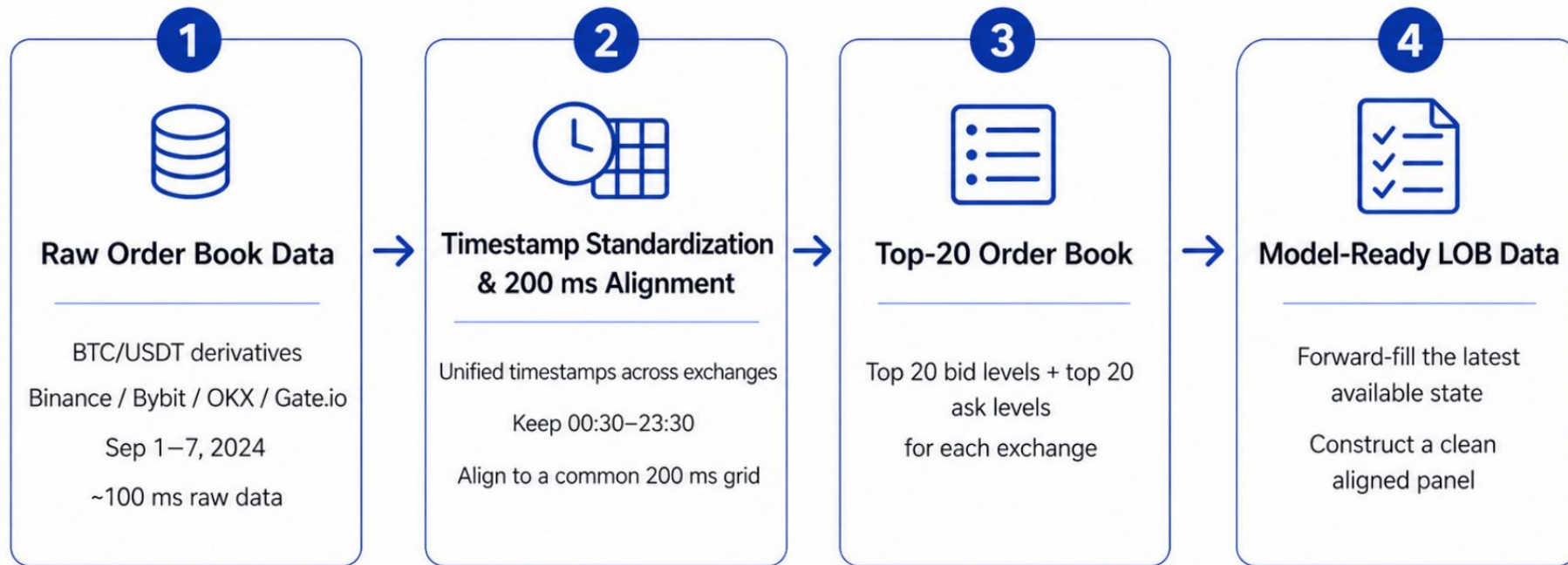
- ❑ Three class classification result (Up, Down and Flat)
- ❑ Five class classification result (Add Confident Up and Confident Down)
- ❑ Build models that incorporate only single exchange signals
- ❑ Build models that incorporate single exchange signals and cross exchange signals
- ❑ Model comparison
- ❑ Discussion result



Data Overview

- i. Data Source & Preprocessing
- ii. Label Design
- iii. Feature Engineering

Data Source & Preprocessing



Label Design



Regression

5 seconds forward
return of midprice

$$y_t = \log \left(\frac{m_{t+5s}}{m_t} \right)$$

Signed Label

$$L_t^{\text{sign}} = \begin{cases} +1, & y_t > 0 \\ 0, & y_t = 0 \\ -1, & y_t < 0 \end{cases}$$

Threshold Label

$$L_t^{\text{thresh}} = \begin{cases} +1, & y_t > \theta \\ 0, & |y_t| \leq \theta \\ -1, & y_t < -\theta \end{cases}$$

Volatility Scaled Label

$$\sigma_t = \sqrt{\frac{1}{W-1} \sum_{i=1}^W (r_{t-i} - \bar{r}_t)^2} \quad L_t^{\text{vol}} = \begin{cases} +1, & y_t > c \sigma_t \\ 0, & |y_t| \leq c \sigma_t \\ -1, & y_t < -c \sigma_t \end{cases}$$

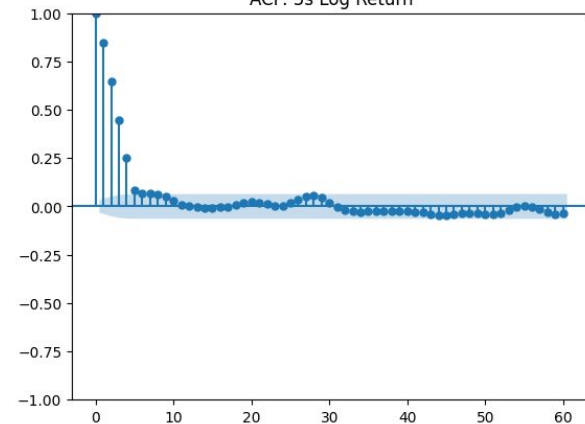
Exploratory Data Analysis



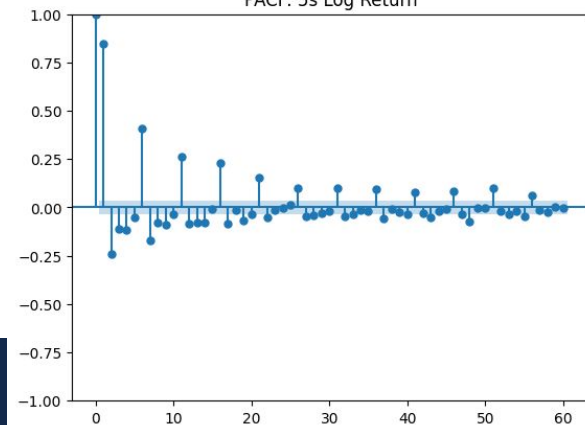
Aligned BTCUSDT Prices Across Exchanges (Bybit, Binance, GateIO, OKX)



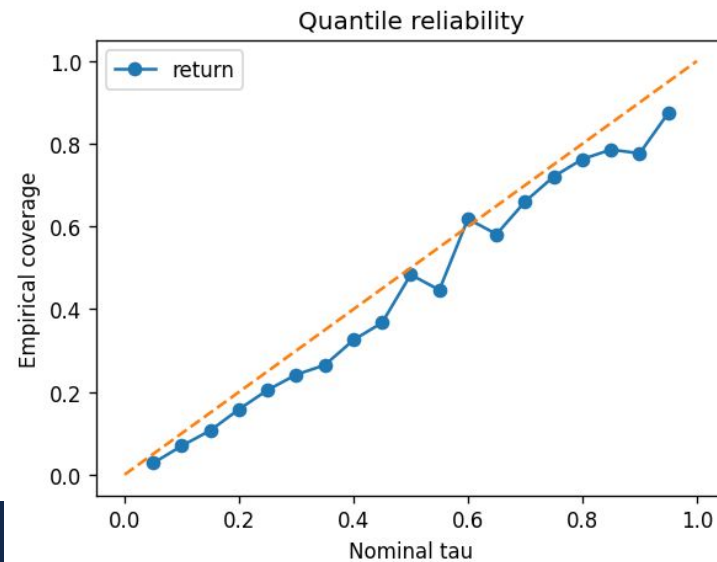
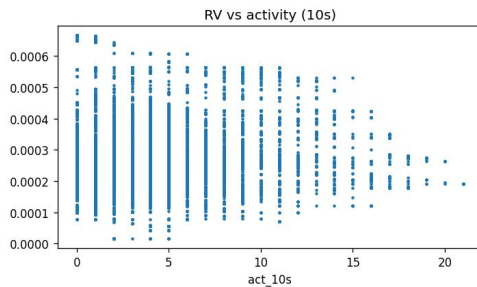
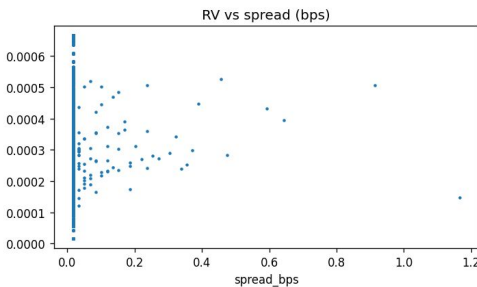
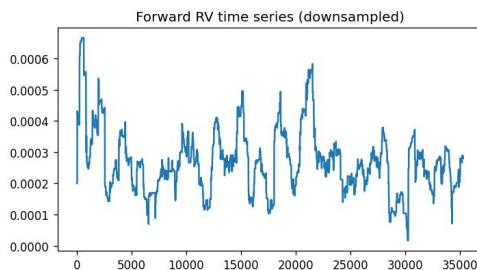
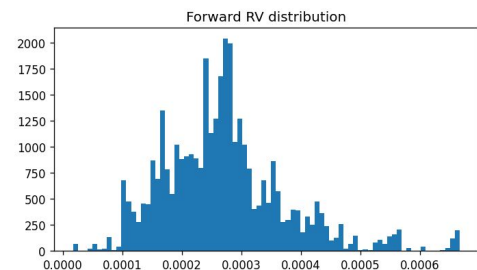
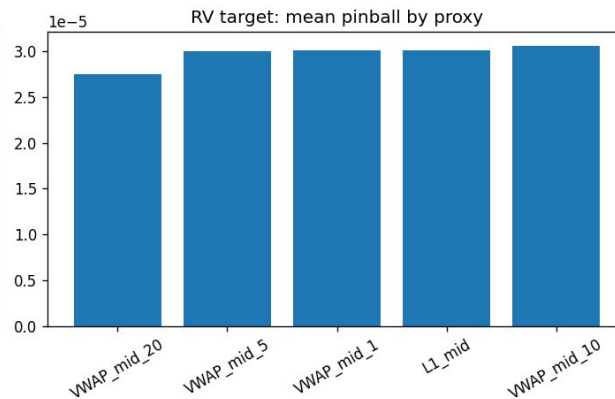
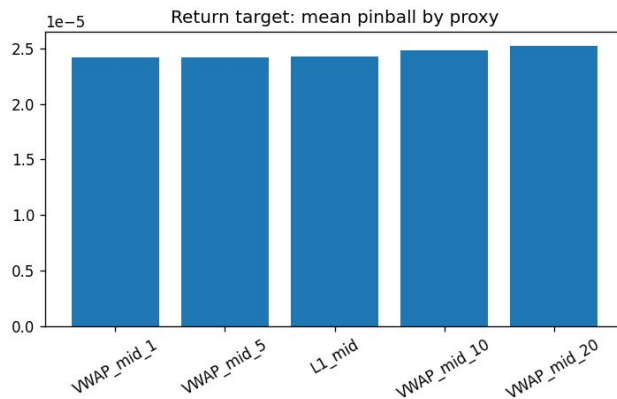
ACF: 5s Log Return



PACF: 5s Log Return



Pinball



Single-Exchange Features



$$\text{L1 Imbalance}_t = \frac{q_{t,1}^b - q_{t,1}^a}{q_{t,1}^b + q_{t,1}^a}$$

Feature Family	What It Measures	Representative Features
Liquidity Pressure	Net buy-vs-sell liquidity pressure across top and deeper book levels	L1 imbalance; cumulative imbalance; notional imbalance; mean level imbalance; microprice skew
Book Geometry	Liquidity concentration, book shape, and price-weighted structure	VWAP bias; top-5 share asymmetry; depth slope; curvature; mean gap
Queue Dynamics	Dynamic changes in displayed bid and ask queues over time	OFI; MLOFI; rolling OFI; quote change; queue depletion
Quote State	Current quote condition, mainly used as control or state variables	mid; spread; relative spread; log spread; micro-minus-mid; VWAP-mid
Market Regime	Short-term volatility and auxiliary market-context variables	realized volatility; jump indicator; MACD; RSI; Bollinger width; time-of-day

$$\text{OFI}_t = \Delta Q_{t,1}^b - \Delta Q_{t,1}^a$$

Single-Exchange LOB Features



Group	Feature Type	What It Measures	Example Features
A	Top-of-book state	Immediate best bid / ask pressure and top-level changes	mid, spread, L1 imbalance, microprice skew, queue change
B	Depth imbalance	Liquidity imbalance across deeper book levels	cumulative imbalance 5 / 10 / 20
C	VWAP / Notional pressure	Price-weighted and dollar-weighted liquidity pressure	VWAP bias, notional imbalance
D	Book shape	Shape, concentration, and slope of liquidity	top-5 share, depth slope, curvature
E	Level imbalance	Imbalance at each individual depth level	L2–L10 imbalance, mean L1–L10 imbalance
F	Order flow imbalance	Dynamic changes in bid / ask queues over time	OFI, MLOFI, rolling OFI 1s / 5s / 10s
G	Returns / Volatility	Recent market movement and volatility regime	short return, realized volatility, jump, momentum
H	Technical / Time context	Broader short-term market context	RSI, MACD, Bollinger width, time-of-day

$$\text{L1 Imbalance}_t = \frac{q_{t,1}^b - q_{t,1}^a}{q_{t,1}^b + q_{t,1}^a}$$

$$\text{Depth Imbalance}_t^{(K)} = \frac{\sum_{i=1}^K q_{t,i}^b - \sum_{i=1}^K q_{t,i}^a}{\sum_{i=1}^K q_{t,i}^b + \sum_{i=1}^K q_{t,i}^a}$$

$$\text{OFI}_t = \Delta Q_{t,1}^b - \Delta Q_{t,1}^a$$

$$\text{RV}_{t,H} = \sqrt{\sum_{j=1}^H r_{t-j}^2}$$

Single-Exchange LOB Features



A. Top-of-book pressure

- Mid price, spread
- L1 imbalance

$$\text{L1 Imbalance}_t = \frac{q_{t,1}^b - q_{t,1}^a}{q_{t,1}^b + q_{t,1}^a}$$

B. Depth imbalance

$$\text{Depth Imbalance}_t^{(K)} = \frac{\sum_{i=1}^K q_{t,i}^b - \sum_{i=1}^K q_{t,i}^a}{\sum_{i=1}^K q_{t,i}^b + \sum_{i=1}^K q_{t,i}^a}$$

- 5 / 10 / 20-level cumulative imbalance

C. Price-weighted depth

- VWAP deviation from mid
- Notional imbalance

D. Book shape

- Liquidity concentration
- Price elasticity
- Curvature

E. Per-level imbalance profile

- Level 2–10 imbalance

F. Order flow imbalance

- Best-level / Multi-level OFI
- Cross-level OFI slope

$$\text{OFI}_t = \Delta Q_{t,1}^b - \Delta Q_{t,1}^a$$

G. Returns and volatility state

- Realized volatility
- Momentum
- Jump indicator
- Volatility regime ratio

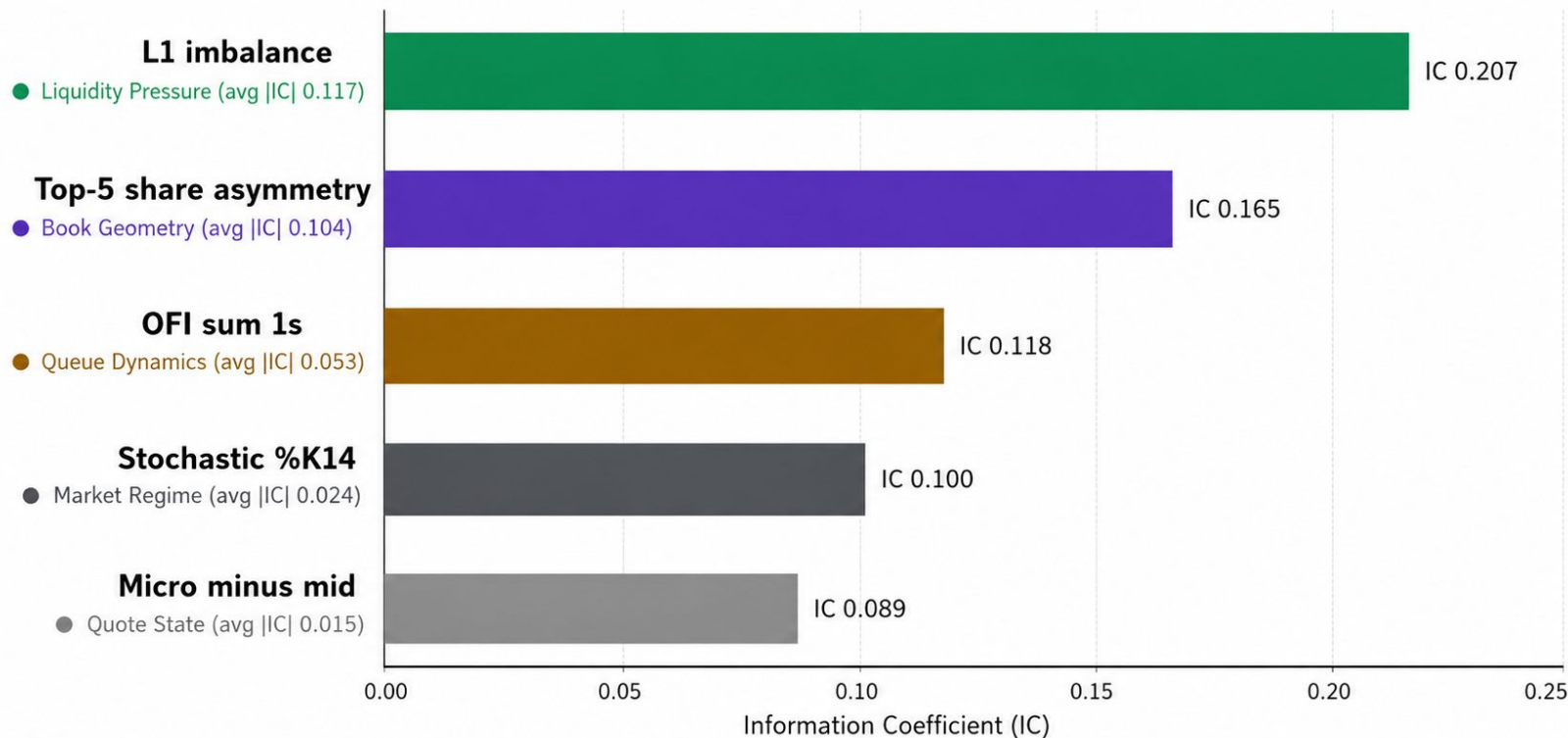
$$\text{RV}_{t,H} = \sqrt{\sum_{j=1}^H r_{t-j}^2}$$

H. Technical and seasonal context

- MACD histogram, RSI
- Bollinger bandwidth, Stochastic %K

Single-Exchange Features

$$IC = \text{Corr}(f_t, y_{t+25})$$



Cross-Exchange Features



Raw source-venue signals: 1-bar return, spread (bps), and L1 imbalance from each non-target exchange

Lag expansion: each signal is expanded over multiple lags to capture delayed information transmission

Co-movement: rolling cross-exchange correlations over short and long windows

Relative pricing: log price spread between source and target exchanges

Lead-lag structure: weighted centroid of cross-correlation across lags to identify which venue tends to move first

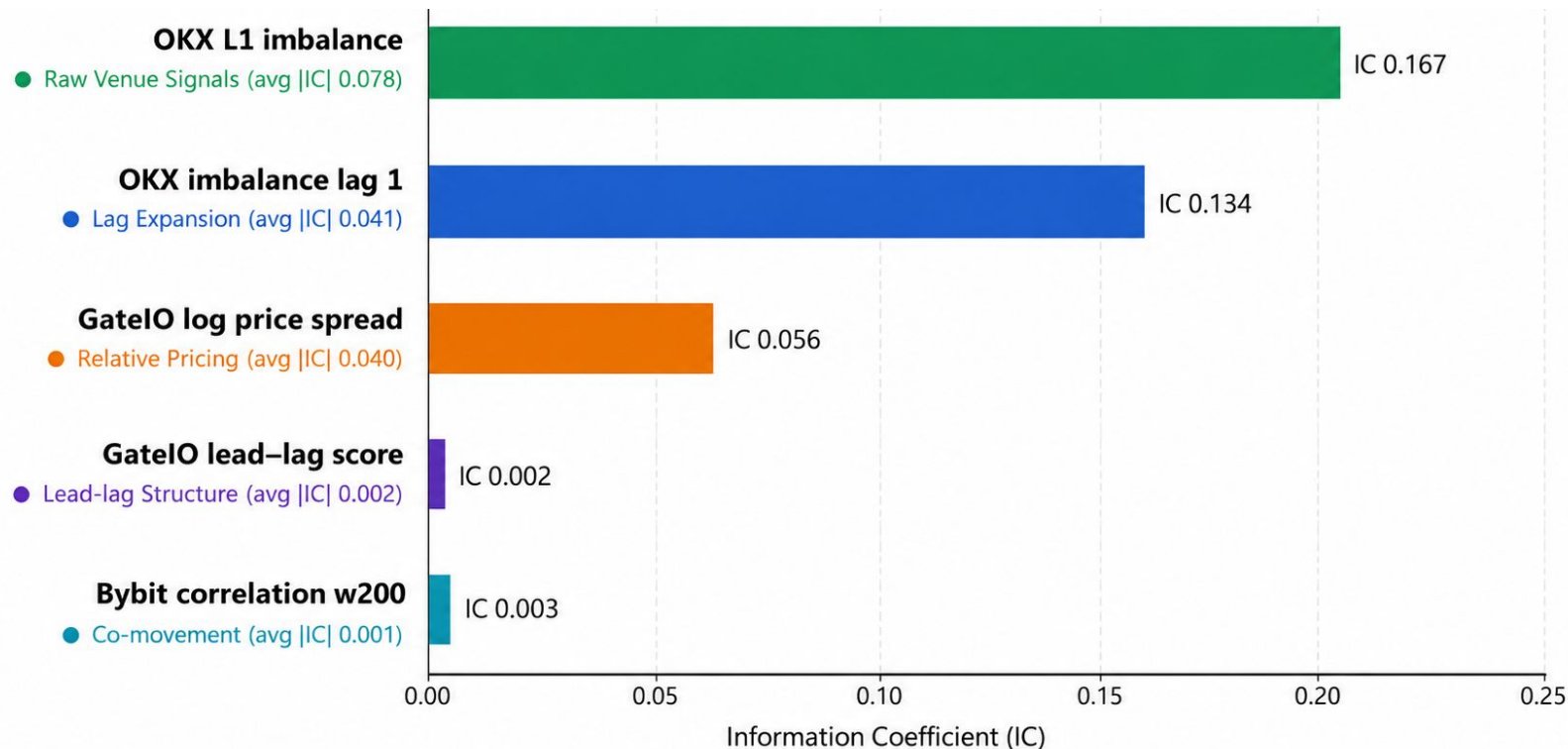
Overall, these features let us move beyond a single-book view and explicitly model information propagation, temporary dislocations, and lead-lag relationships across exchanges.

$$\text{LeadLag}^{(src \rightarrow tgt)} = \frac{\sum_{k=0}^K k |\rho_k|}{\sum_{k=0}^K |\rho_k|}$$

$$\rho_k = \text{Corr}\left(r_t^{(tgt)}, r_{t-k}^{(src)}\right), \quad k = 0, \dots, K$$

Cross-Exchange Features

$$IC = \text{Corr}(f_t, y_{t+25})$$





Models

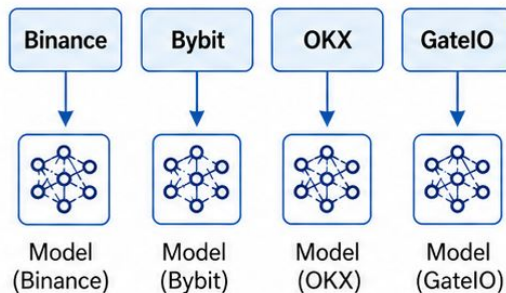
- i. XGBoost
- ii. CatBoost
- iii. LightGBM
- iv. Random Forest
- v. GRU
- vi. Hawkes Process

Modeling Setups

Three setups, total = 9 models

Setup 1

Single-Exchange Only

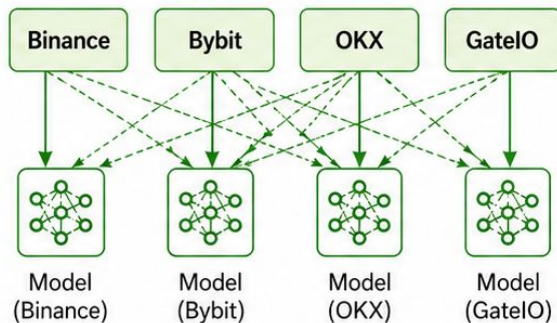


- **Input:** local features only
- **Prediction:** separate per exchange
- **Models:** 4

4 models

Setup 2

Single-Exchange + Cross-Exchange Separate Prediction

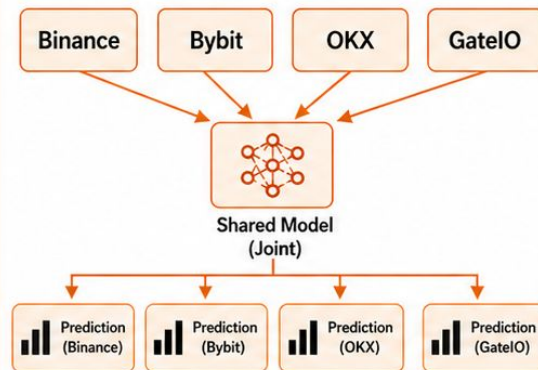


- **Input:** local + cross-exchange features
- **Prediction:** separate per exchange
- **Models:** 4

4 models

Setup 3

Single-Exchange + Cross-Exchange Joint Prediction



- **Input:** local + cross-exchange features
- **Prediction:** all 4 exchanges together
- **Models:** 1

1 model

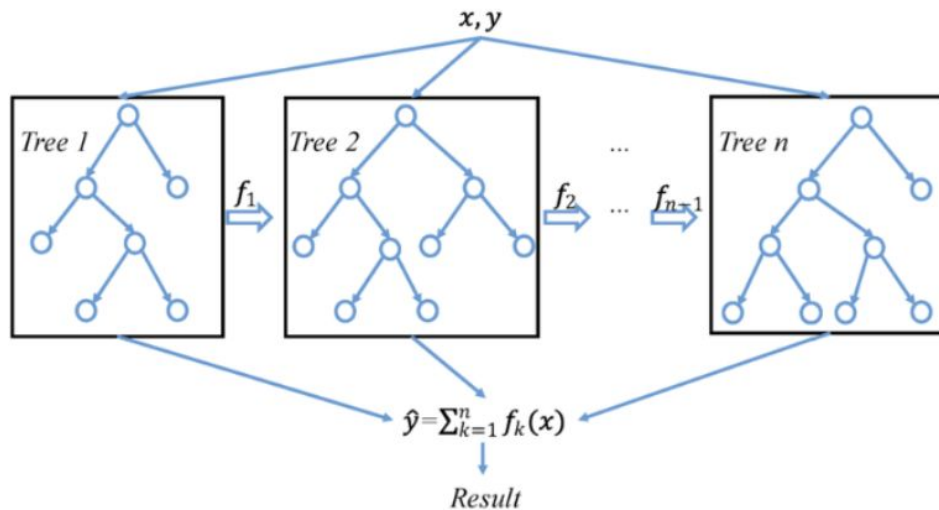
Total models = 4 + 4 + 1 = 9

Key comparison: local-only vs. local+cross features, and separate vs. joint prediction.

Baseline Model: XGBoost



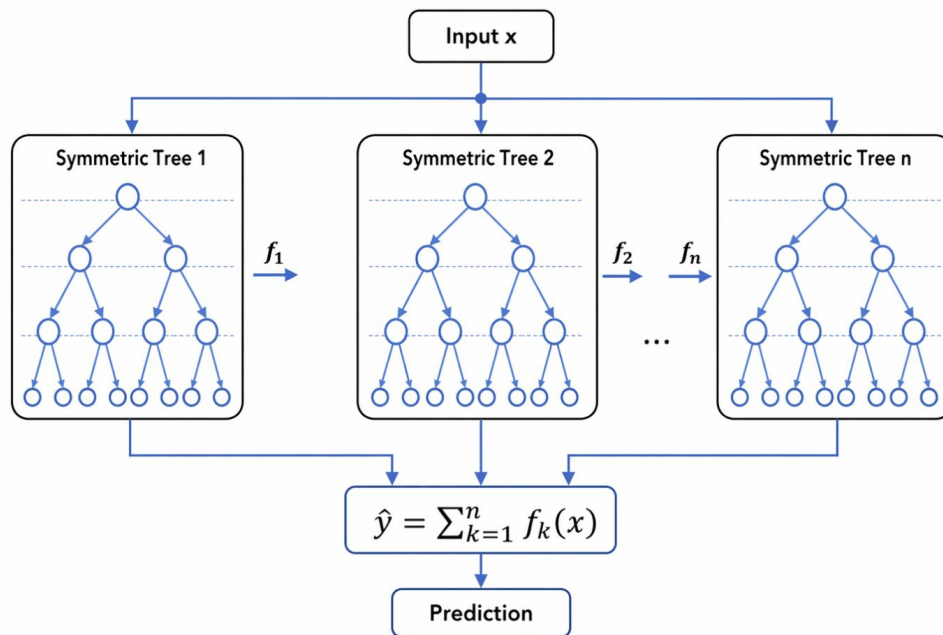
- Ensemble tree model based on gradient boosting
 - Trees are added sequentially
 - Each tree reduces previous prediction errors
- Main strengths
 - Captures nonlinear effects
 - Learns feature interactions automatically
 - Supports regularization for better generalization
- Why relevant here
 - LOB features are high-dimensional and structured
 - Cross-exchange signals may be nonlinear



CatBoost



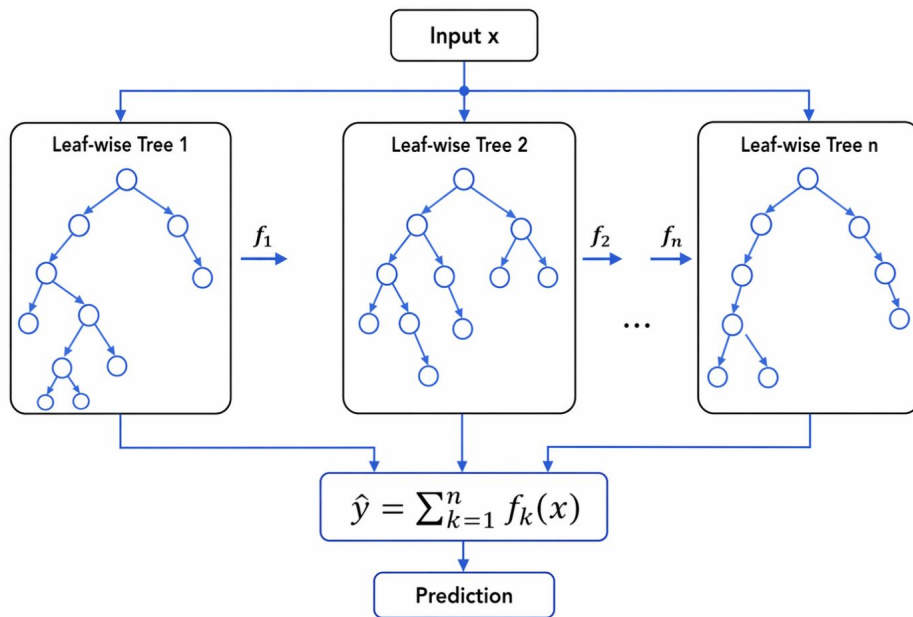
- Gradient boosting tree model
- Builds trees sequentially to reduce errors
- Uses symmetric trees for efficient training



LightGBM



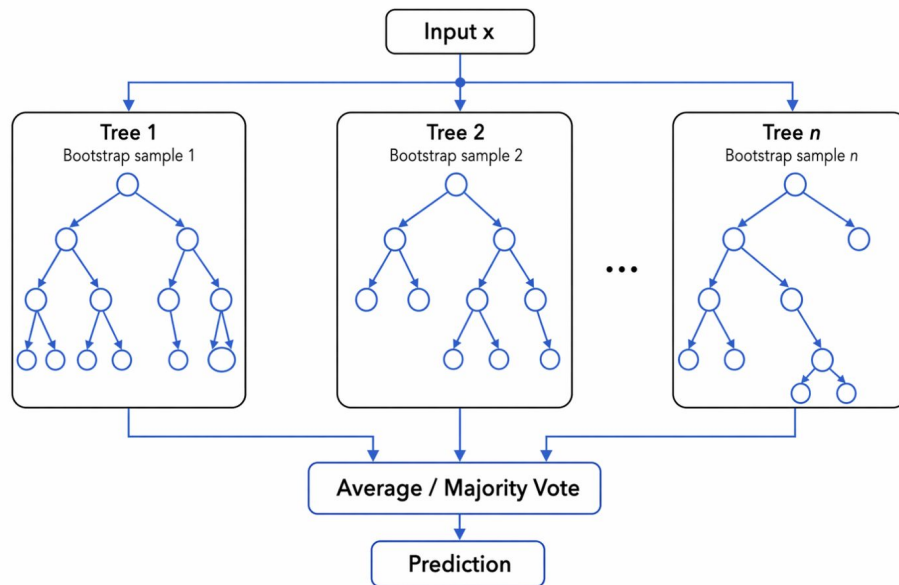
- Gradient boosting tree model
- Uses leaf-wise growth rather than symmetric trees
- Faster training on large, high-dimensional datasets



Random Forest



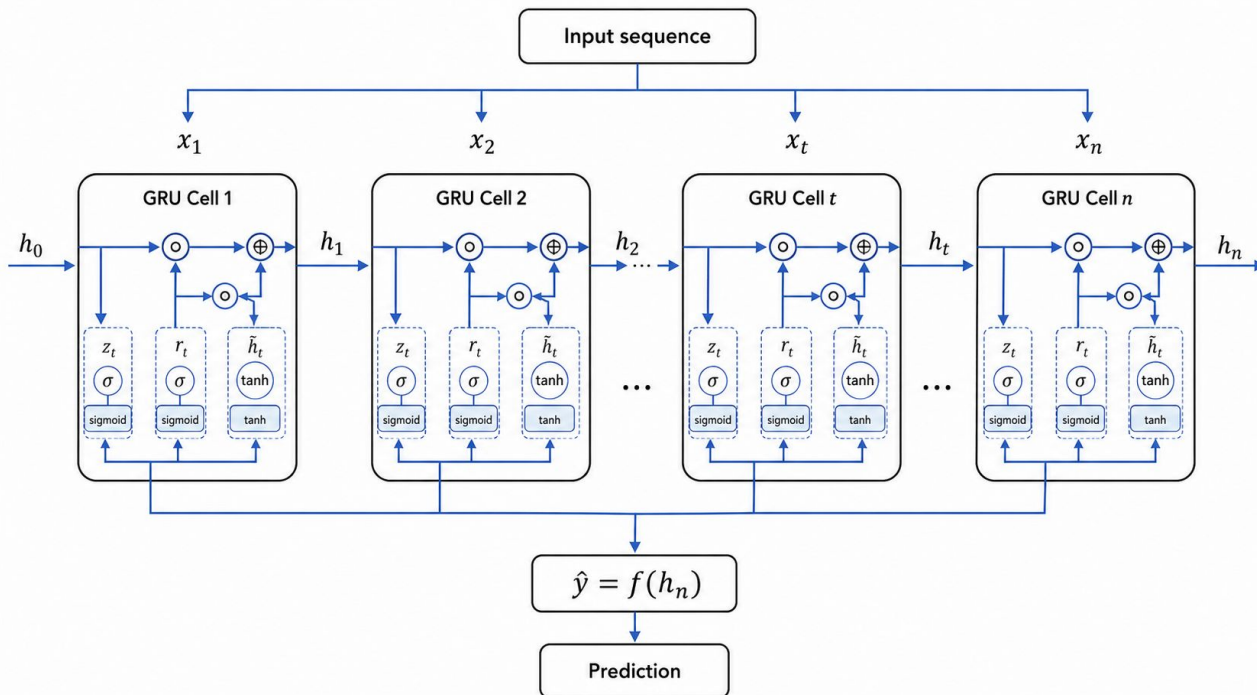
- Ensemble model of decision trees
- Trains trees in parallel rather than sequentially
- Improves stability by averaging multiple trees



Gated Recurrent Unit (GRU)



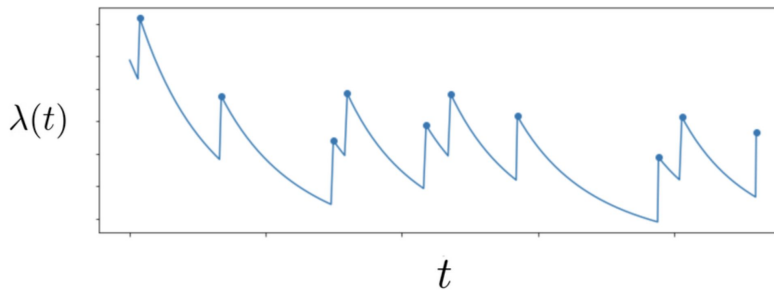
- Deep learning sequence model
- Uses recent LOB feature sequence, not one static snapshot
- Hidden state summarizes recent market dynamics
- Gates decide what information to keep or forget



Hawkes Process



$$\lambda(t | H_t) = s(t) + \sum_{t_i < t} \phi(t - t_i)$$



- ❑ Self-exciting nature
- ❑ Intensity Function: Given the known history, the intensity of event happening at time t
- ❑ Baseline Intensity: The intensity of event happening at time t neglecting all history events
- ❑ Kernel Function: The influence of an event happened at time t_i on the current event happening intensity

Hawkes Process



- Hawkes Intensities (k represents a specific event):

$$\lambda_t^{(k)} = e^{-\beta_k \Delta t} \lambda_{t-1}^{(k)} + E_{t-1}^{(k)}$$

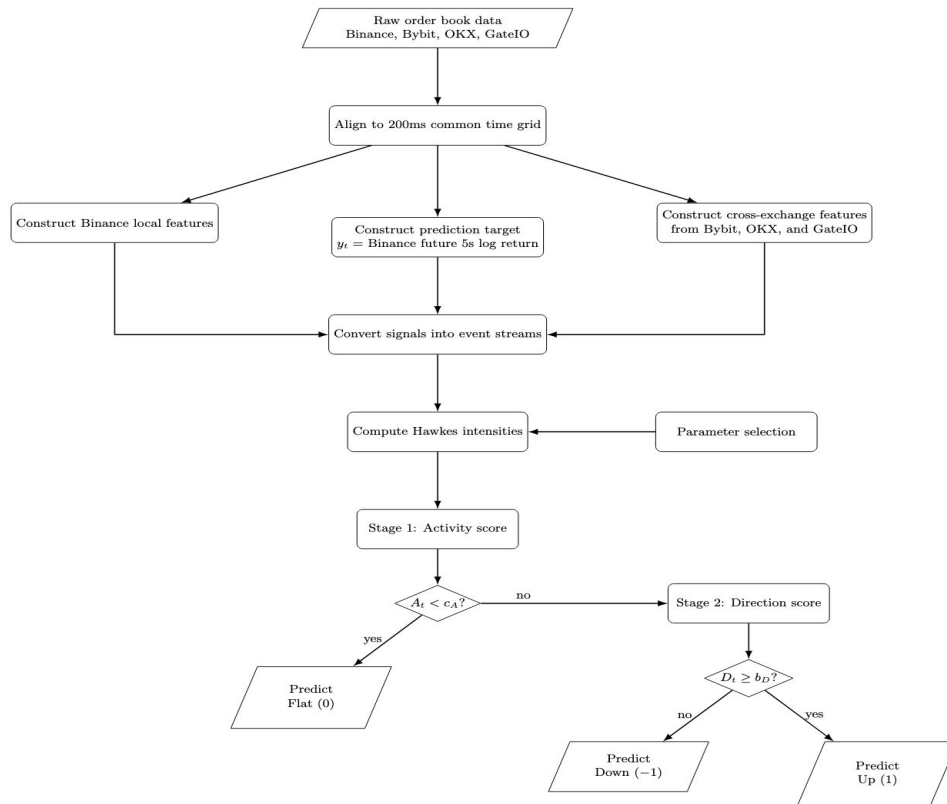
$$E_{t-1} = \begin{cases} 1, & \text{if the event occurs at time } t-1, \\ 0, & \text{otherwise.} \end{cases}$$

- Activity Score:

$$A_t = \lambda_t^{\text{up}} + \lambda_t^{\text{down}} + \lambda_t^{\text{buy}} + \lambda_t^{\text{sell}} \\ + \lambda_t^{\text{bybit,up}} + \lambda_t^{\text{bybit,down}} + \lambda_t^{\text{okx,up}} + \lambda_t^{\text{okx,down}} \\ + \lambda_t^{\text{gateio,up}} + \lambda_t^{\text{gateio,down}}.$$

- Direction Score:

$$D_t = \lambda_t^{\text{up}} - \lambda_t^{\text{down}} + \lambda_t^{\text{buy}} - \lambda_t^{\text{sell}} \\ + \lambda_t^{\text{bybit,up}} - \lambda_t^{\text{bybit,down}} + \lambda_t^{\text{okx,up}} - \lambda_t^{\text{okx,down}} \\ + \lambda_t^{\text{gateio,up}} - \lambda_t^{\text{gateio,down}}.$$





Result

- i. Single-Exchange Modeling
- ii. Cross-Exchange Modeling

Baseline Model: XGBoost



Headline metrics

Accuracy: 51.61%

Macro F1: 0.508

Weighted F1: 0.512

Spearman IC: 0.259 (t=243.7)

$\delta=0.00$: directional accuracy **45.99%**

$\delta=0.30$: **59.46%**

$\delta=0.40$: **63.84%**

$\delta=0.50$: **68.25%**

As we require stronger class separation, directional accuracy increases substantially, although coverage falls from 100% to 7.9%.

Top features by XGBoost gain

- D_slope_asym_5: Book shape asymmetry
- A_micro_skew: Microprice skew
- C_notional_ask_5: Notional ask depth
- A_micro_minus_mid: Microprice deviation from mid
- imb_vol_adj: Vol-adjusted imbalance

XGBoost



max_depth	learning rate = 0.05		learning rate = 0.10		learning rate = 0.20	
	subsample = 0.8	subsample = 1.0	subsample = 0.8	subsample = 1.0	subsample = 0.8	subsample = 1.0
3	0.5083 (231)	0.5084 (242)	0.5035 (175)	0.5047 (124)	0.5068 (69)	0.5026 (48)
5	0.5062 (157)	0.5059 (147)	0.5064 (87)	0.5014 (86)	0.5023 (39)	0.4962 (33)
7	0.5021 (111)	0.5019 (112)	0.4985 (60)	0.4964 (48)	0.4797 (32)	0.4859 (23)

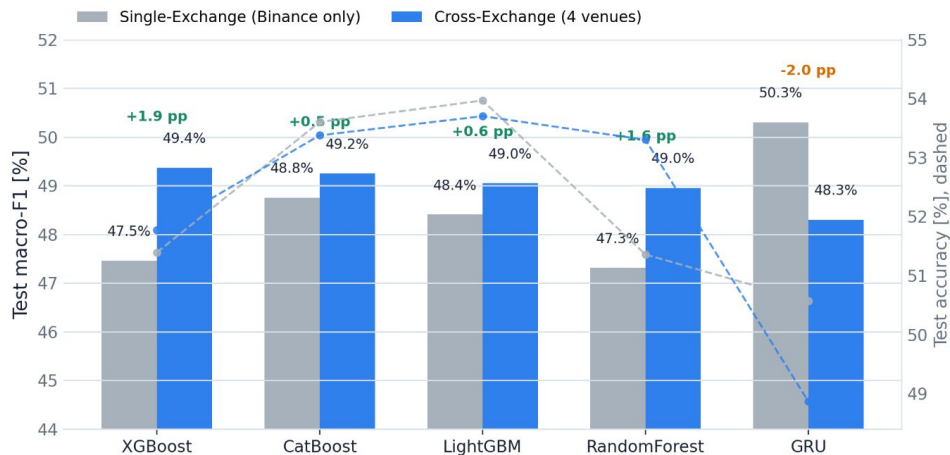
Entries report Macro F1, with the corresponding early-stopping round in parentheses.

Cross-Exchange features improve prediction



Cross-Exchange Features Improve Balanced Direction Classification

Bars show test macro-F1; dashed markers show test accuracy on the same train/validation/test split.



Main takeaway

Cross-Exchange wins macro-F1 in 4/5 models

Average macro-F1 gain:
+0.54 pp

Average accuracy change:
+0.03 pp

Accuracy is mixed; macro-F1
better reflects balanced
Down/Flat/Up performance.

Figure. Cross-Exchange features improve macro-F1 for the tree models; GRU is added as a neural baseline. Dashed markers show accuracy for context; accuracy can move differently when Flat labels dominate the test set.

Prediction result: 3 class confusion matrix



Three-Class Test Confusion Matrices

Rows are true Down/Flat/Up labels; columns are model predictions. Cell color and percent are row-normalized.

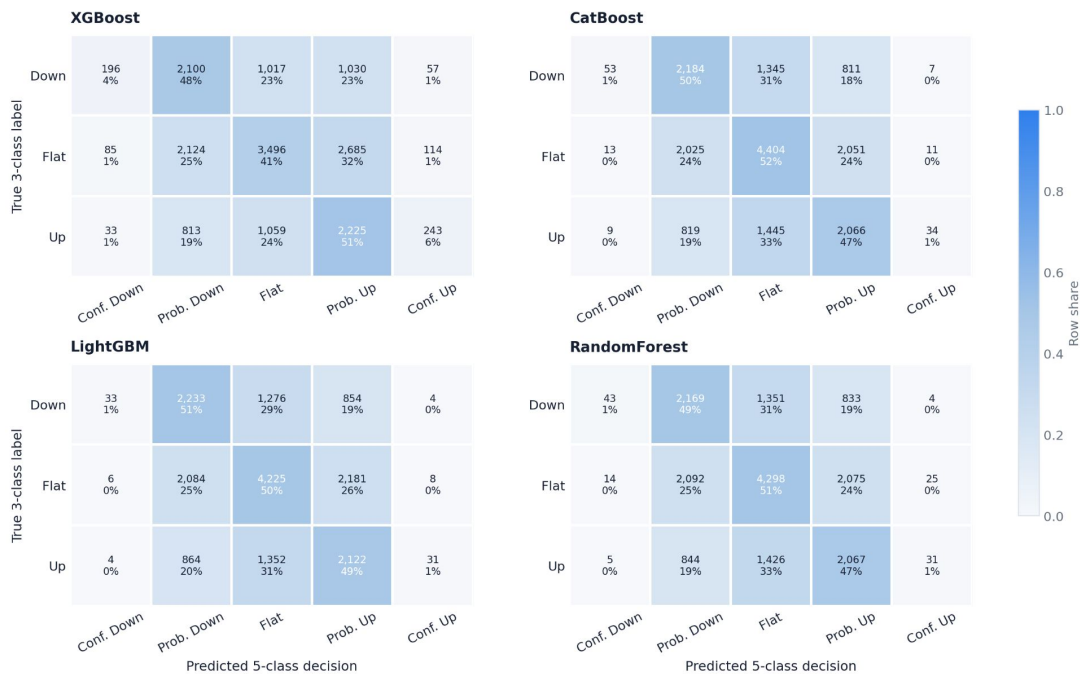


Prediction result: 5 class confusion matrix



Five-Class Decision Confusion Matrices

Rows use the available true Down/Flat/Up labels; columns show threshold-based five-class decisions. Cell percent is row-normalized.

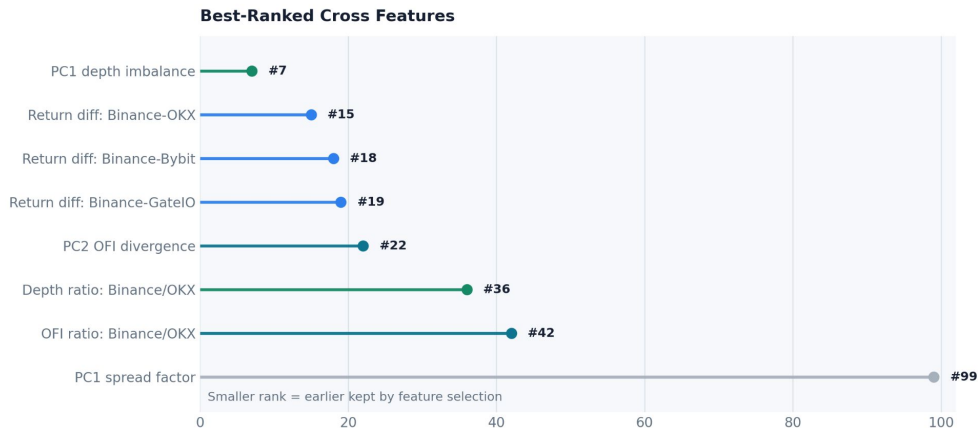


Which Cross-Exchange Signals Are Most Important?



Best-Ranked Cross-Exchange Features

Earlier selection rank means the feature contributed sooner to the top-100 set.



Takeaway: the earliest cross features are the shared depth-imbalance factor, Binance-vs-peer return gaps, and OFI/depth ratios. The spread factor only appears very late.

$$ReturnDiff_{Binance,OKX,t} = r_{Binance,t}^{(h)} - r_{OKX,t}^{(h)}$$

$$DI_{e,t} = \frac{\text{bid depth} - \text{ask depth}}{\text{bid depth} + \text{ask depth}}$$

$$PC1DepthImbalance_t = w_1 DI_{Binance,t} + w_2 DI_{OKX,t} + w_3 DI_{Bybit,t} + w_4 DI_{GateIO,t}$$

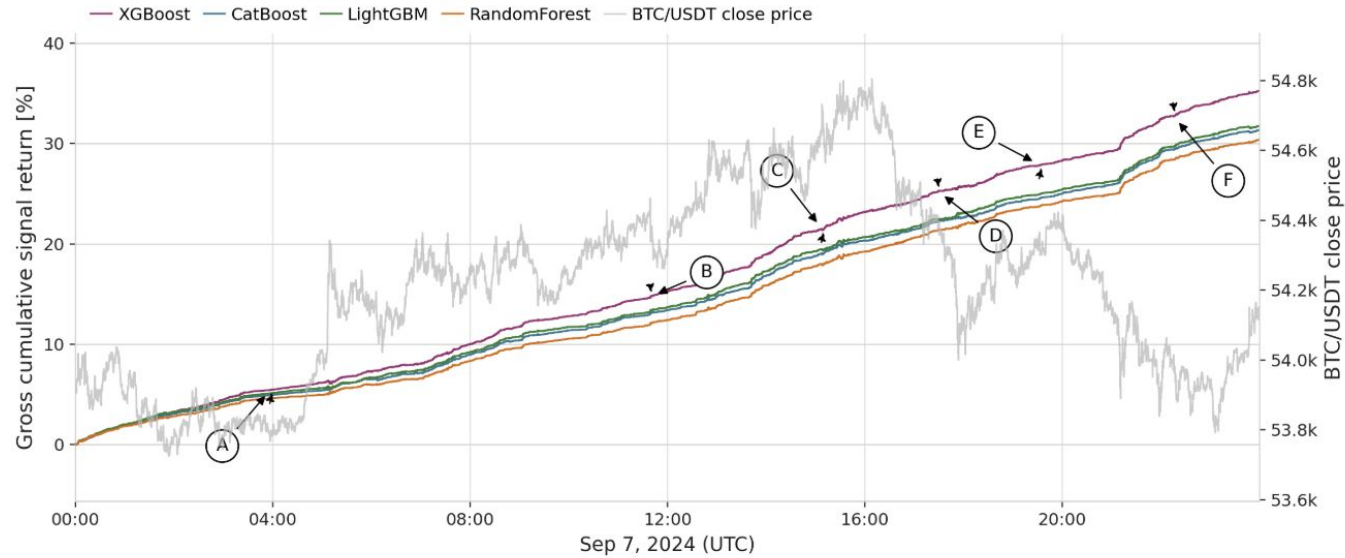


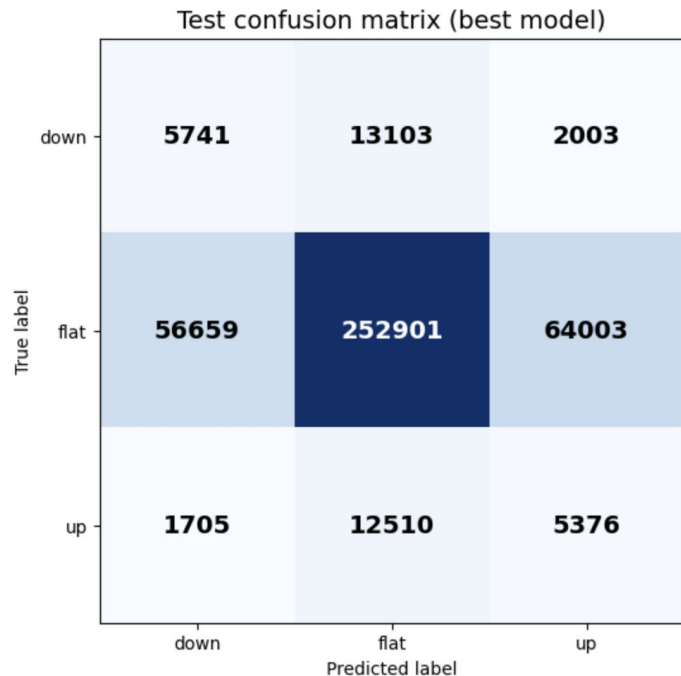
Figure. Gross zero-cost cumulative signal return for four tree-model 5-second trading signals on the held-out test day. BTC/USDT close price is shown in the background for market context; labelled circles mark large intraday price moves, and dashed arrows indicate the nearby strategy response. Use classification metrics, not this gross curve alone, for model ranking.



4bps RT, enter every signal

Horizon	F1 (up/dn)	Prec UP	Prec DN	Note	Horizon	n_trades	hit%	net PnL
5s	0.1482	0.102	0.072	very sparse true positives	5s	13,093	60.4%	-\$42,248
10s	0.2076	0.165	0.104		10s	7,264	61.0%	-\$23,112
30s	0.2820	0.234	0.187		30s	3,275	57.8%	-\$10,051
60s	0.3233	0.256	0.271		60s	2,018	55.7%	-\$5,933
120s	0.3329	0.300	0.338	best F1	120s	1,010	52.9%	-\$3,470
300s	0.2913	0.370	0.337		300s	409	50.1%	-\$2,005

Hawkes Process Result Visualization



- ❑ When the predicted label is up, 97.2% of the time the actual label is up or flat
- ❑ When the predicted label is down, 97.3% of the time the actual label is down or flat
- ❑ Better at identifying tranquil market periods
- ❑ There are some useful insights



Discussion

i. Limitations

Limitations

- ❑ Limited training data and computational resources
- ❑ Forward-filling missing values during preprocessing may introduce misleading information
- ❑ High-frequency crypto data is noisy
- ❑ Current pipeline is too slow for real-time deployment

Future Works

- ❑ Train on longer time periods and more market conditions
- ❑ Apply Kalman filter on the original data and compare results
- ❑ Train more diverse models, including reinforcement learning
- ❑ Use high-memory CPU servers and cloud computing
- ❑ Include missingness indicators to our model





Thank You



Please feel free to ask questions.

Appendix: Feature Evaluation Framework



200ms bars | label = 5s forward log return

Metric	Formula	Interpretation / Rough Guide
IC	$IC = \text{Corr}(f_t, y_{t+25})$	Predictive strength. >0.10 strong; 0.05 – 0.10 useful.
ICIR	$ICIR = \frac{\text{mean}(IC_d)}{\text{std}(IC_d)}$	Cross-day stability. >2 good; >3 strong; >5 very stable.
t-stat	$t = ICIR \sqrt{N}$	Significance of average IC. $ t > 2$ significant; $ t > 3$ strong.
QA mono	$\text{Mono}(Q_1, \dots, Q_5)$	Quantile monotonicity. 1.0 ideal; ≥ 0.8 good; ≈ 0.5 noisy.
ACF1	$ACF_1 = \text{Corr}(f_t, f_{t-1})$	One-lag persistence. Too close to 1 may be slow/non-stationary.
Half-life	$HL = \frac{\log(0.5)}{\log(ACF_1)}$	Signal decay speed. 4 bars = 0.8 s on a 200 ms grid.

Note: thresholds are rough guides for this high-frequency setting; overlapping 5s labels mean t-stat is supporting evidence, not final proof of tradable alpha.

Appendix

Group	Feature	Definition / Formula	Intuition
A	Mid price	$m_t = \frac{a_1 + b_1}{2}$	Reference price of the book
	Spread	$s_t = a_1 - b_1$	Trading cost / liquidity tightness
	Relative spread	$\frac{a_1 - b_1}{m_t}$	Spread normalized by price level
	L1 imbalance	$\frac{q_1^b - q_1^a}{q_1^b + q_1^a}$	Whether top-level liquidity is bid-heavy or ask-heavy
	Microprice	$\frac{a_1 q_1^b + b_1 q_1^a}{q_1^b + q_1^a}$	Size-weighted fair price; moves toward the thinner side
	Microprice skew	$\frac{\text{microprice} - m_t}{s_t}$	Normalized top-book pressure
B	Cumulative depth imbalance	$\frac{\sum_{i=1}^L q_i^b - \sum_{i=1}^L q_i^a}{\sum_{i=1}^L q_i^b + \sum_{i=1}^L q_i^a}$	Bid/ask pressure across deeper levels
C	Notional imbalance	$\frac{\sum_{i=1}^L b_i q_i^b - \sum_{i=1}^L a_i q_i^a}{\sum_{i=1}^L b_i q_i^b + \sum_{i=1}^L a_i q_i^a}$	Dollar-weighted liquidity imbalance
	VWAP bias	$\frac{\text{VWAP}_L - m_t}{m_t}$	Whether volume-weighted book price is above or below mid
D	Top-5 share	$\frac{\sum_{i=1}^5 q_i}{\sum_{i=1}^{20} q_i}$	Liquidity concentration near the top
	Shape asymmetry	Bid shape – Ask shape	Whether liquidity is more concentrated on bid or ask side
	Slope / curvature	Fitted slope or curvature of depth over levels	Whether the book is steep, flat, convex, or concave
E	Per-level imbalance	$\frac{q_i^b - q_i^a}{q_i^b + q_i^a}$	Imbalance at each individual level
	Mean L1–L10 imbalance	$\frac{1}{10} \sum_{i=1}^{10} \frac{q_i^b - q_i^a}{q_i^b + q_i^a}$	Average imbalance across the first 10 levels

Appendix

Group	Feature	Definition / Formula	Intuition
A / F	Queue change	$\Delta q_{1,t}^b = q_{1,t}^b - q_{1,t-1}^b, \Delta q_{1,t}^a = q_{1,t}^a - q_{1,t-1}^a$	Immediate change in best bid / ask queue
A / F	Queue depletion rate	$\frac{(q_{1,t-1}^b - q_{1,t}^b)^+}{q_{1,t-1}^b + \epsilon}$	How fast top-level liquidity disappears
F	Order Flow Imbalance	$OFI_t = \Delta q_{1,t}^b - \Delta q_{1,t}^a$	Dynamic buying vs selling pressure
F	Normalized OFI	$\frac{OFI_t}{q_{1,t}^b + q_{1,t}^a + \epsilon}$	OFI adjusted by top-book size
F	Rolling OFI	$\sum_{u=W+1}^t OFI_u, W = 1s, 5s, 10s$	Recent accumulated order-flow pressure
F	Multi-level OFI	$MLOFI_{i,t} = \Delta q_{i,t}^b - \Delta q_{i,t}^a$	Order-flow imbalance at deeper levels
F	OFI slope	Slope of $MLOFI_{i,t}$ across levels	Whether order flow is concentrated near the top
G	Grid return	$r_t = \log\left(\frac{m_t}{m_{t-1}}\right)$	Most recent midprice return
G	Realized volatility	$\sqrt{\sum_{u=t-W+1}^t r_u^2}$	Recent short-term volatility
G	Jump indicator	$\mathbf{1}\{ r_t > k\sigma_t\}$	Whether recent return is unusually large
G	Momentum	$\sum_{u=t-W+1}^t r_u$	Recent directional price movement
G	Vol-adjusted imbalance	$\frac{imbalance_t}{\sigma_t + \epsilon}$	Imbalance scaled by current volatility regime
H	MACD histogram	$MACD - Signal$	Short-term trend / reversal signal
H	RSI	Relative strength index over recent returns	Overbought / oversold pressure
H	Bollinger width	$\frac{Upper\ band - Lower\ band}{Moving\ average}$	Volatility expansion or compression
H	Time-of-day	$\sin(2\pi \tau), \cos(2\pi \tau)$	Intraday periodicity



	SINGLE	CROSS_OLD	CROSS_MATE
macro_f1	0.503	0.478	0.455
f1_down	0.502	0.507	0.508
f1_flat	0.478	0.389	0.326
f1_up	0.529	0.537	0.533
IC_spearman	0.341	0.350	0.343
n_feat	233	329	585

Confusion Matrix (test)

=== SINGLE ===

	DOWN	FLAT	UP
true_DOWN	64066	24957	33112
true_FLAT	43202	67097	53754
true_UP	25886	24560	77272

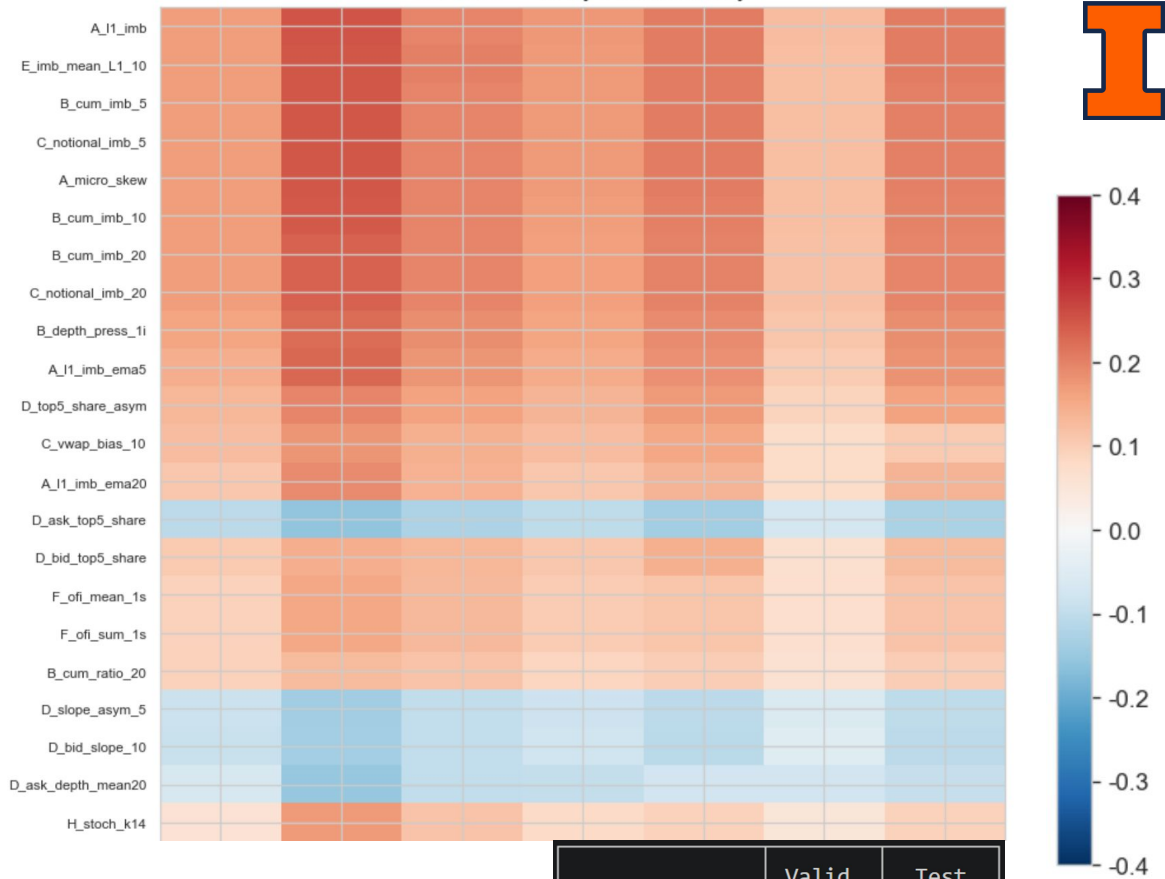
=== CROSS_OLD ===

	DOWN	FLAT	UP
true_DOWN	67281	16328	38526
true_FLAT	48976	47101	67976
true_UP	26974	14700	86044

=== CROSS_MATE ===

	DOWN	FLAT	UP
true_DOWN	68856	12587	40692
true_FLAT	51985	36480	75588
true_UP	28322	10861	88535

Binance — IC by Feature x Day



	Valid	Test
IC (Spearman)	0.1956	0.3439
Macro F1	0.4713	0.5030



I ILLINOIS